Optimal Skew Swap Replication

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Abstract

As a follow-up to *Optimal Variance Swap Replication* [3], we propose a payoff function for realized skew swap. We then show the proposed realized skew formula can be decomposed into difference of variance and gamma swaps both of which can be replicated using vanilla options. Finally, we use the Gaussian quadrature method to conclude that one needs approximately eight vanilla options to closely replicate a six-month skew swap exposure for SPX.

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Introduction

As follow-up to [3], we propose a payoff function for realized skew swap which also matches the one proposed by [2]. We then show the proposed realized skew formula can be decomposed into difference of variance and gamma swaps both of which can be replicated using vanilla options as shown in [4]. Finally, we use the Gaussian quadrature method of [3] to conclude that one needs approximately eight vanilla options to closely replicate a six-month skew swap exposure for SPX. Our contribution is two-fold. First, we give a definition of skew swap payoff that is intuitive as it is for a variance swap as opposed to esoteric decomposition as difference of variance and gamma swap hence can be more marketable. Secondly, we use methods of [3] to replicate/hedge such a product.

Skew Swap

We propose skew swap payoff given by the following formula

$$S - K_s, \qquad S = -\frac{1}{T}\sum_{i=1}^{n}\sum_{j=1}^{i} r_i^2 r_j \tag{1}$$

where $r_i = ln\left(\frac{S_{t_i}}{S_{t_{i-1}}}\right)$, n is number of returns, and K_s is strike of the skew swap. Using 252-day convention, one has $T = \frac{n}{252}$. Here, we note that we are taking negative of usual skewness formula so that positive S means stock return has left-skewed distribution. Not only can this payoff be closely replicated using variance and gamma swaps but its expected value in absence of arbitrage can be shown to correspond to skewness of the terminal stock distribution. Both results motivated the proposal for skew swap formula in equation (1). Let's define skewness as third moment of the terminal distribution of underlying such that

$$skew = ln^3 \frac{S_T}{S_0}$$

Expected value of $skew$ is then

$$E[skew] = \sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n} E\left[r_i r_j r_k\right]$$

Now in the absence of arbitrage (alternatively as dictated by efficient market hypothesis), any future return should be unforecastable, hence

$$E\left[r_i r_j r_k\right] = 0 \qquad \text{if } i \neq j \neq k \tag{2}$$

$$E\left[r_i^2 r_j\right] = 0 \qquad \text{if } j > i \tag{3}$$

Equation (2) and (3) just state that in the absence of arbitrage, autocorrelation of future stock returns to past measurable events should be zero. On the other hand, spot moves drive subsequent instantaneous volatility $E[r_i^2]$ as seen by for example, leverage effect [1]. Hence sum of $E\left[r_i^2 r_j\right]$ in equation (1) for $j \leq i$ capture both the leverage and the long-memory volatility clustering effects discussed in [6, 7].

Putting this together, one has

$$E[skew] = 3\sum_{i=1}^{n}\sum_{j=1}^{i-1} E\left[r_i^2 r_j\right] + \sum_{i=1}^{n} E[r_i^3]$$

Up to diagonal term, this is exactly the definition of above skew payoff formula, hence one has

$$E[S] \cong -\frac{1}{3T} E[skew] \tag{4}$$

Skew Swap Replication

In this section, we note how equation (1) can be well-approximated by using variance and gamma swap contracts. Payoff of a variance swap is given as

$$V = \frac{1}{T}\sum_{i=1}^{n} r_i^2$$

Similarly, payoff of gamma swap is

$$\Sigma = \frac{1}{T}\sum_{i=1}^{n}\left(\frac{S_{t_i}}{S_{t_0}}\right) r_i^2$$

Taking the difference of two,

$$V - \Sigma = \frac{1}{T}\sum_{i=1}^{n}\left(1 - \frac{S_{t_i}}{S_{t_0}}\right) r_i^2$$

Using the fact that $\frac{S_{t_i}}{S_0} = exp\left(\sum_{j=1}^{i} r_j\right)$, above can be expressed as

$$V - \Sigma = \frac{1}{T}\sum_{i=1}^{n}\left(1 - exp\left(\sum_{j=1}^{i} r_j\right)\right) r_i^2$$

To leading order,

$$V - \Sigma = -\frac{1}{T}\sum_{i=1}^{n}\sum_{j=1}^{i} r_i^2 r_j \cong S \tag{5}$$

Hence, we can replicate and price skew swaps by taking equal units of variance and gamma swaps, which each of these contracts can be replicated by vanilla options using well-known results, see for example [4]. We furthermore note that realized skewness replication of equation (5) is consistent with results derived in [2]. Below, we plot realized skew defined in equation (1) for different rolling observation windows.

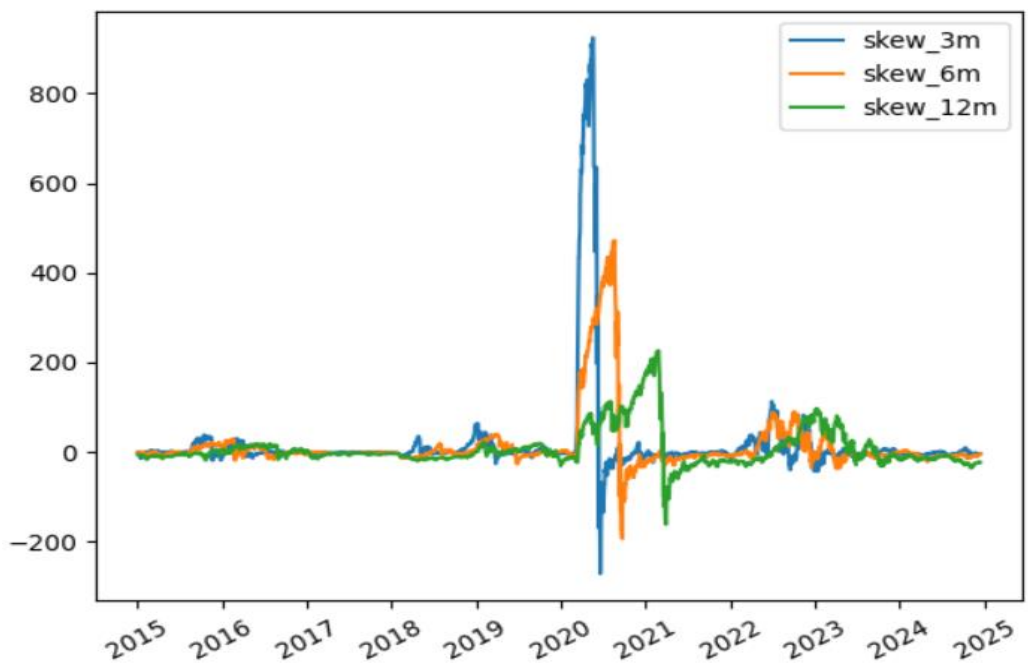


Fig 1. 3, 6, 12 month realized skew: $-\frac{10000}{n} 252 \sum_{i=1}^{n}\sum_{j=1}^{i} r_i^2 r_j$ for 2015-2024 period

As expected, realized skew experienced large shocks during COVID and during brief bear market of 2022 as well as stock market declines in late 2018 and China devaluation in 2015. Below, we show three-month realized skew and similarly three-month realized volatility since 2022.

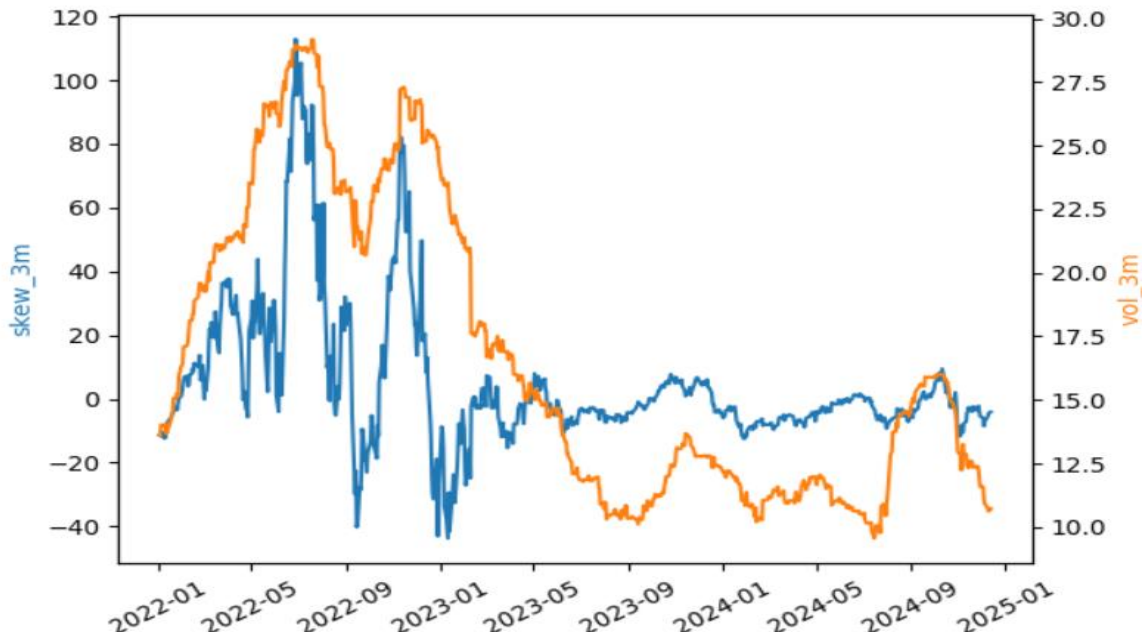


Fig 2. 3 month realized skew and volatility for 2022-2024 period

Lastly, we plot the approximations used in equation (5) for a six month realized skew to demonstrate that variance-gamma formula does an excellent job of approximating the payoff even during extreme periods seen in COVID.

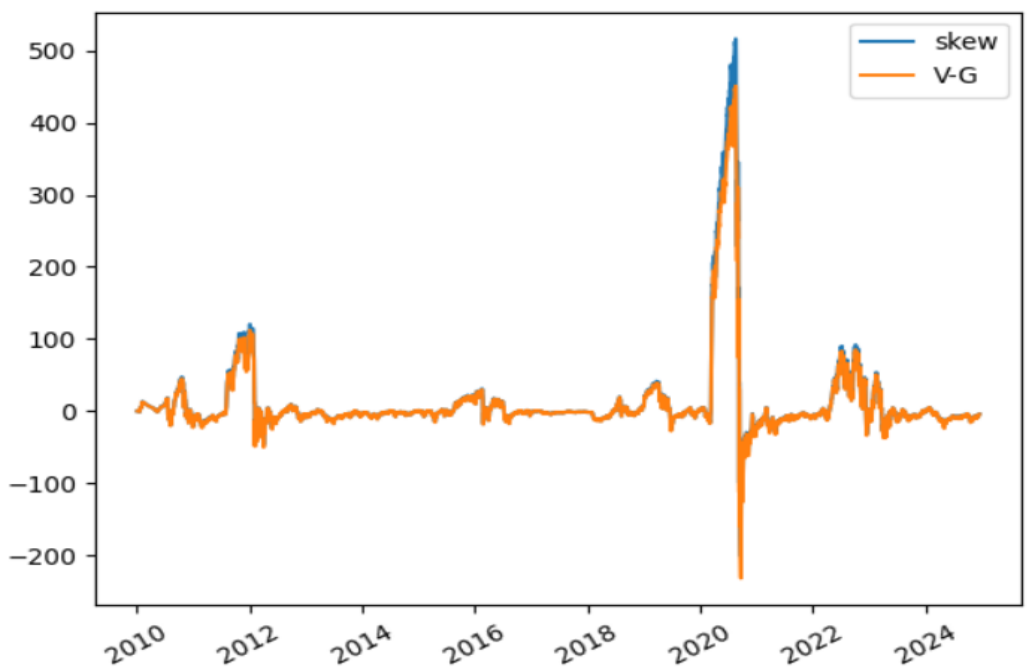


Fig 3. 6 month realized skew formula versus variance – gamma swap payoff

Pricing of Skew Swap

To get the fair value of skew swap strike, we set $E[S - K_s] = 0$ and solve for fair strike K_s such that

$$K_s = E[S] \cong E[V] - E[\Sigma]$$

Using the well-known replication procedure (see for example [3]), one has

$$E[V] = \frac{2}{T}\left(\int_0^{S_0} put(S_0, K)\frac{1}{K^2}dK + \int_{S_0}^{\infty} call(S_0, K)\frac{1}{K^2}dK\right)$$

$$E[\Sigma] = \frac{2}{TS_0}\left(\int_0^{S_0} put(S_0, K)\frac{1}{K}dK + \int_{S_0}^{\infty} call(S_0, K)\frac{1}{K}dK\right)$$

Finally, one has

$$K_s = \frac{2}{TS_0}\left(\int_0^{S_0} put(S_0, K)\frac{1}{K^2}(S_0 - K)dK + \int_{S_0}^{\infty} call(S_0, K)\frac{1}{K^2}(S_0 - K)dK\right) \quad (6)$$

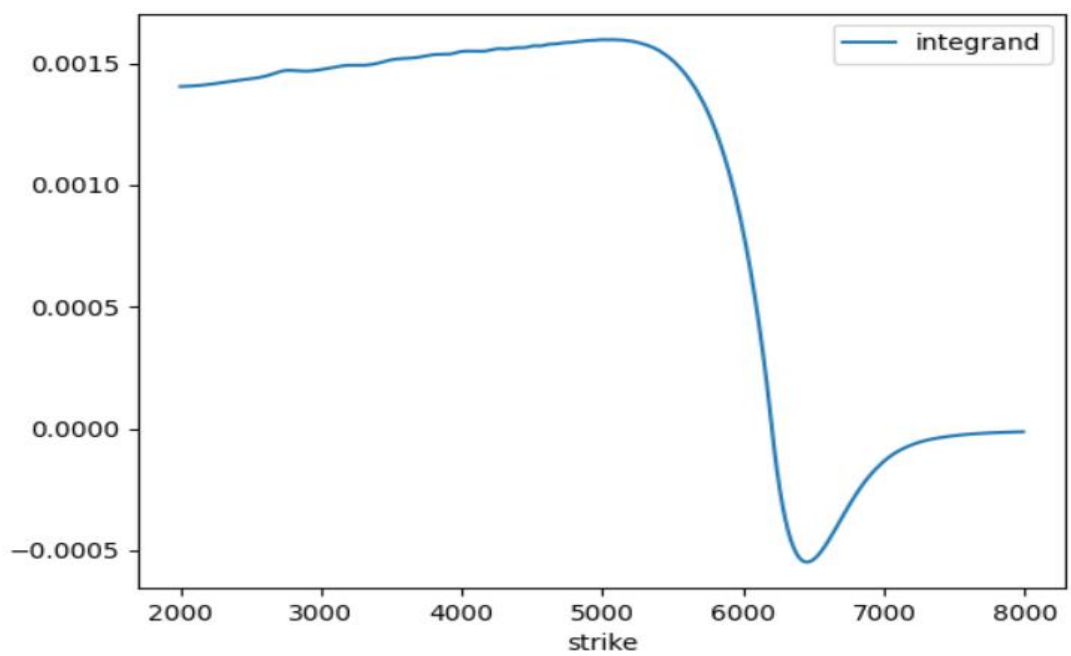


Fig 4. May 16, 25 SPXW skew swap integrand as of Dec 6, 2024 close

We find in practice that proper extrapolation of implied vol surface, especially for puts, has greater impact on skew swaps than on either the variance swaps or gamma swaps.

Gaussian-Quadrature Replication

Following the results of [3] we apply Gaussian-Quadrature methods to replicate above skew swap payoff. For the put side, we have

$$K_s^{put} = \frac{2}{TS_0}\int_0^{S_0} put(S_0,K)\frac{1}{K^2}(S_0-K)dK \cong \frac{S_0-K_{min}}{TS_0}\sum_{i=1}^{n} w_i \frac{(S_0-K_i)}{K_i^2} put(S_0,K_i) \quad (7)$$

where

$$K_i = \frac{1}{2}\left(S_0 + K_{min} + \xi_i(S_0 - K_{min})\right) \quad (8)$$

$$w_i = \frac{2}{(1-\xi_i^2)[P_n'(\xi_i)]^2} \quad (9)$$

and ξ_i are i^{th} roots of the Legendre polynomials, $P_n(x)$, with n being the number of put options used.

For the call side, similarly one has

$$K_s^{call} = \frac{2}{TS_0}\int_{S_0}^{\infty} call(S_0,K)\frac{1}{K^2}(S_0-K)dK \cong \frac{K_{max}-S_0}{TS_0}\sum_{j=1}^{m} w_j \frac{(S_0-K_j)}{K_j^2} call(S_0,K_j) \quad (10)$$

where

$$K_j = \frac{1}{2}\left(S_0 + K_{max} + \xi_j(K_{max} - S_0)\right) \quad (11)$$

and m is the number of call options used. Here we do not make change of variable $x = \frac{1}{K}$ as in [3] since below graph shows the change of variable does not help for skew swaps.

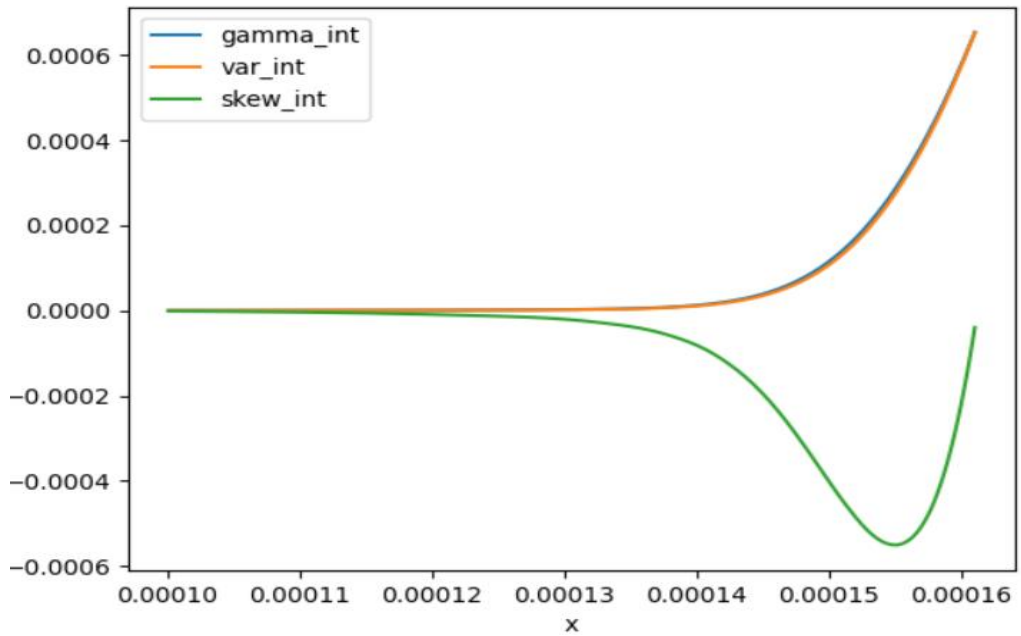


Fig 5. May 16, 2025 SPXW call integrand as of Dec 6, 2024 close for gamma, var and skew swaps vs $\frac{1}{K}$

Using the results from [5], we have following roots and weights for the Gauss-Legendre quadrature.

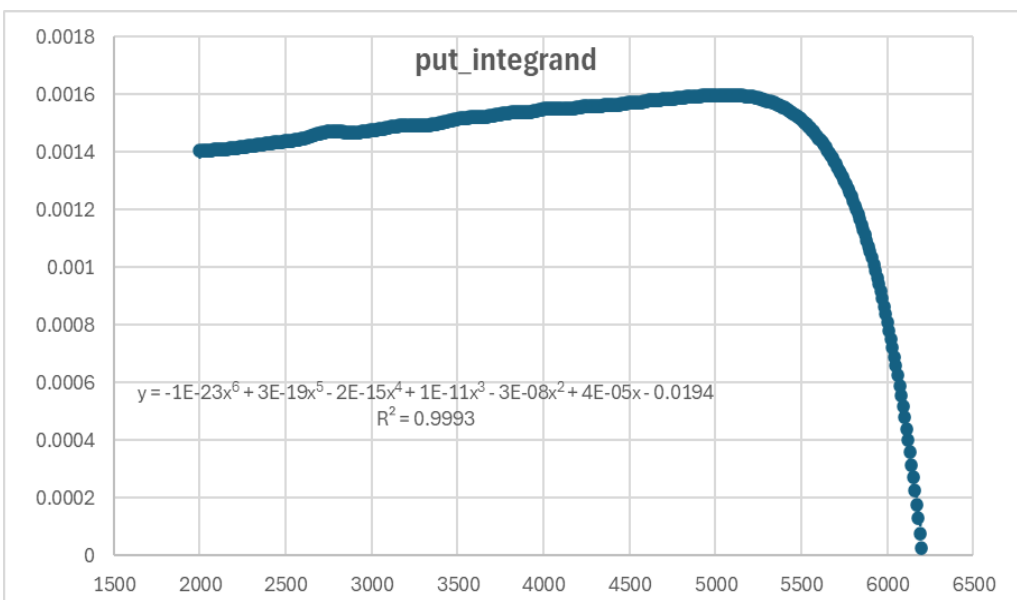
Number of points, n	Points, x_i		Weights, w_i	
1	0		2	
2	$\pm\frac{1}{\sqrt{3}}$	±0.57735...	1	
3	0		$\frac{8}{9}$	0.888889...
	$\pm\sqrt{\frac{3}{5}}$	±0.774597...	$\frac{5}{9}$	0.555556...
4	$\pm\sqrt{\frac{3}{7}-\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.339981...	$\frac{18+\sqrt{30}}{36}$	0.652145...
	$\pm\sqrt{\frac{3}{7}+\frac{2}{7}\sqrt{\frac{6}{5}}}$	±0.861136...	$\frac{18-\sqrt{30}}{36}$	0.347855...
5	0		$\frac{128}{225}$	0.568889...
	$\pm\frac{1}{3}\sqrt{5-2\sqrt{\frac{10}{7}}}$	±0.538469...	$\frac{322+13\sqrt{70}}{900}$	0.478629...
	$\pm\frac{1}{3}\sqrt{5+2\sqrt{\frac{10}{7}}}$	±0.90618...	$\frac{322-13\sqrt{70}}{900}$	0.236927...

Below we summarize the main results of this section

Table 1. Skew Swap Replicating Vanilla Option Portfolio

Replication Procedure	**Puts ($i = 1$ to n)**	**Calls ($j = 1$ to m)**
Number of options	$w_i \frac{(S_0 - K_{min})}{TS_0} \frac{(S_0 - K_i)}{K_i^2}$	$w_j \frac{(K_{max} - S_0)}{TS_0} \frac{(S_0 - K_j)}{K_j^2}$
Strikes to use	$\frac{1}{2}\left(S_0 + K_{min} + \xi_i(S_0 - K_{min})\right)$	$K_j = \frac{1}{2}\left(S_0 + K_{max} + \xi_j(K_{max} - S_0)\right)$

Finally, we fit polynomials to the skew swap integrand to solve for optimal number of replicating vanilla options.



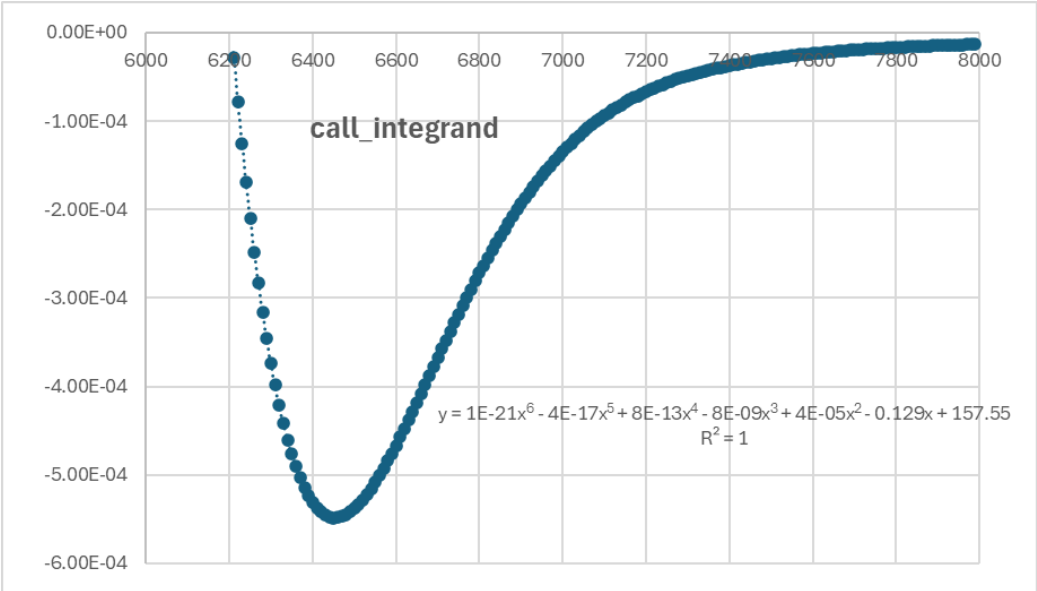


Fig 6. Polynomial fits to May 16, 2025 skew swap put and call integrand as of Dec 6, 2024 close

Using results of [3], we see that we need approximately (6+1)/2 ~ 4 puts and 4 calls to closely replicate May 16, 2025 SPXW skew swap as of Dec 6, 2024.

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